

Syndrome Controlled Adaptive Sliding Window Decoding

Minimal modification of the current staircase decoder

1 Objective

The current decoding schedule is kept unchanged. For a target block $\mathbf{B}_i$, the decoder still performs forward sweeps through the component constraints in the current window. The proposed modification adds only two mechanisms:

- an early stopping test based on the syndrome of the target component constraint;
- an increase of the window size when the ordinary sweep budget is exhausted and the target syndrome is still nonzero.

No backward sweep, reliability threshold, stagnation counter, or additional message type is introduced.

Core rule. After every complete forward sweep, check the syndrome associated with the target block. If it is zero, release the block. If it is nonzero, continue with another sweep. If it remains nonzero after the available sweeps, enlarge the window and continue.

2 Parameters

Let w_0 be the window size already used by the decoder. The following values are recommended for the first implementation.

Parameter	Meaning	Initial choice
w_0	Ordinary window size	current value of w
$w_{\max}$	Largest permitted window size	$w_0 + 2$
Δw	Window increment	1
V_i	Maximum number of sweeps at one window size	$v_{I,\max}$ for $i = 1$, otherwise $v_{\max}$
I_{SPA}	Maximum internal SPA iterations for one component row	current <code>spa_max_iter</code>

The possible window sizes are therefore

$$w_0, \quad w_0 + 1, \quad w_0 + 2.$$

All existing values of α_c , α_n , $v_{\max}$, $v_{I,\max}$, and `spa_max_iter` remain unchanged. The purpose of the first experiment is to isolate the effect of syndrome controlled stopping and window enlargement.

3 Syndrome checked after a sweep

When the target block is $\mathbf{B}_i$, the relevant component constraint is

$$\mathbf{A}_i = \left[\mathbf{B}_{i-1}^\top \mid \mathbf{B}_i \right].$$

The syndrome must be checked **after the complete forward sweep has ended**. It must not be checked immediately after decoding $\mathbf{A}_i$, because the later constraints in the same sweep can still modify the LLRs of $\mathbf{B}_i$.

At the end of the sweep, obtain the hard decision for every entry of $\mathbf{B}_i$ from

$$\mathbf{L}_{\text{app}}(\mathbf{B}_i) = \mathbf{L}_{I,i} + \mathbf{L}_{E,i}.$$

For each entry (a, b) , use

$$\hat{B}_i(a, b) = \begin{cases} 1, & L_{\text{app}}(B_i(a, b)) < 0, \\ 0, & L_{\text{app}}(B_i(a, b)) \geq 0. \end{cases}$$

For $\mathbf{B}_{i-1}$, use the hard decision that was previously accepted and released by the decoder. Denote it by $\hat{\mathbf{B}}_{i-1}^{\text{acc}}$. Then form

$$\hat{\mathbf{A}}_i = \left[\left(\hat{\mathbf{B}}_{i-1}^{\text{acc}} \right)^\top \mid \hat{\mathbf{B}}_i \right].$$

For each row $k = 1, \dots, m$, compute

$$\mathbf{s}_{i,k} = \hat{\mathbf{A}}_i(k, :) \mathbf{H}^\top \pmod{2}.$$

The target syndrome is verified when

$$\mathbf{s}_{i,k} = \mathbf{0} \quad \text{for every } k = 1, \dots, m.$$

Equivalently,

$$\hat{\mathbf{A}}_i \mathbf{H}^\top = \mathbf{0}.$$

If the component decoder uses a permuted bit order, each row of $\hat{\mathbf{A}}_i$ must be converted to the same order used by $\mathbf{H}$ before the multiplication. The syndrome must never be computed by mixing the systematic order with the raw order.

It is also useful to record the syndrome weight

$$u_i = \sum_{k=1}^m \text{wt}(\mathbf{s}_{i,k}),$$

where u_i is the total number of unsatisfied parity checks. In the proposed first implementation, u_i is logged for analysis, but the decision uses only whether u_i is zero or nonzero.

4 Adaptive decoding procedure

For every target block $\mathbf{B}_i$, apply the following procedure.

1. Set the current window size to

$$w_{\text{cur}} \leftarrow w_0.$$

2. Set

$$V_i = \begin{cases} v_{I,\text{max}}, & i = 1, \\ v_{\text{max}}, & i > 1. \end{cases}$$

3. Set the local sweep counter to $v \leftarrow 0$.

4. Perform one complete forward sweep through the current window:

$$\mathbf{A}_i \rightarrow \mathbf{A}_{i+1} \rightarrow \cdots \rightarrow \mathbf{A}_{i+w_{\text{cur}}-1}.$$

If the end of the received sequence is closer, stop at the last available component constraint.

5. Increase the local counter: $v \leftarrow v + 1$.

6. Compute the target syndrome of $\hat{\mathbf{A}}_i$ using the procedure in the previous section.

7. If the syndrome is zero, accept $\hat{\mathbf{B}}_i$, store it as $\hat{\mathbf{B}}_i^{\text{acc}}$, extract its information bits, and advance to target block $\mathbf{B}_{i+1}$.

8. If the syndrome is nonzero and $v < V_i$, perform another forward sweep with the same window size and return to Step 5.

9. If the syndrome is nonzero and $v = V_i$, check the current window size.

- If $w_{\text{cur}} < w_{\text{max}}$, enlarge the window according to

$$w_{\text{cur}} \leftarrow \min\{w_{\text{cur}} + \Delta w, w_{\text{max}}\}.$$

Keep all channel and extrinsic LLRs already computed. Reset only the local sweep counter, $v \leftarrow 0$, and return to Step 4.

- If $w_{\text{cur}} = w_{\text{max}}$, declare a decoding failure for $\mathbf{B}_i$.

Important. Enlarging the window does not restart decoding. The existing LLRs are retained. The newly included component constraint is simply added to the next forward sweeps.

5 Decision table

Condition after a complete sweep	Action
Target syndrome is zero	Accept $\mathbf{B}_i$ immediately and advance the window
Target syndrome is nonzero and $v < V_i$	Keep the same window and perform another forward sweep
Target syndrome is nonzero, $v = V_i$, and $w_{\text{cur}} < w_{\text{max}}$	Increase the window by one, retain all messages, reset the local sweep counter
Target syndrome is nonzero, $v = V_i$, and $w_{\text{cur}} = w_{\text{max}}$	Declare a decoding failure for $\mathbf{B}_i$

In compact form,

$u_i = 0$	$\implies$	accept $\mathbf{B}_i$,
$u_i > 0, v < V_i$	$\implies$	perform another sweep,
$u_i > 0, v = V_i, w_{\text{cur}} < w_{\text{max}}$	$\implies$	increase the window,
$u_i > 0, v = V_i, w_{\text{cur}} = w_{\text{max}}$	$\implies$	declare failure.

6 Example

Assume

$$w_0 = 4, \quad w_{\text{max}} = 6, \quad \Delta w = 1, \quad V_i = 5.$$

The decoder first uses the window

$$\mathbf{A}_i, \mathbf{A}_{i+1}, \mathbf{A}_{i+2}, \mathbf{A}_{i+3}.$$

Suppose that the syndrome weight after the five sweeps is

$$12, 7, 4, 2, 2.$$

The syndrome is still nonzero after the available five sweeps. The decoder therefore increases the window size to five:

$$\mathbf{A}_i, \mathbf{A}_{i+1}, \mathbf{A}_{i+2}, \mathbf{A}_{i+3}, \mathbf{A}_{i+4}.$$

All existing LLRs are retained and the local sweep counter is reset. Suppose that the new syndrome weights are

$$2, 1, 0.$$

After the third sweep with the enlarged window, the target syndrome is zero. The decoder accepts $\mathbf{B}_i$ and advances to $\mathbf{B}_{i+1}$. The window size is then initialized again to w_0 for the new target block.

7 Quantities to record in simulations

For every target block, record at least:

- whether the block was accepted or declared failed;
- the window size at acceptance;
- the total number of forward sweeps;
- the sequence of syndrome weights u_i ;
- whether an accepted block was actually correct, when the transmitted bits are available.

At the sequence level, report:

BER, FER, failure probability, average number of sweeps, average window size.

This is necessary because the adaptive decoder should be evaluated both in terms of error performance and additional complexity.

8 Interpretation of a zero syndrome

A zero syndrome means that the current hard estimate satisfies the component parity checks. It does not prove that the estimate is the transmitted codeword, because the decoder may converge to another valid codeword. The syndrome is therefore a practical stopping and failure detection criterion, not a perfect error detector. If a CRC is available, the CRC should remain the final error detection mechanism.

First implementation to test. Keep the present forward schedule. Check the target syndrome after every complete sweep. Stop early when it is zero. If it is still nonzero after the normal sweep budget, increase the window by one, up to $w_0 + 2$. If the syndrome remains nonzero at the maximum window and after the full budget, declare failure.